Exponential Functions: The Variable Is in the Exponent

An exponential function has the form $f(x) = Ca^x$, where $a > 0$, $a \neq 1$, and $C \neq 0$. For $C > 0$, $a > 1$ gives growth; $0 < a < 1$ gives decay. The initial value is $f(0) = C$.

$$f(x) = 2^x, \qquad g(x) = \left(\frac{1}{2}\right)^x.$$

Evaluate each at $x = -2, -1, 0, 1, 2$. Identify growth or decay and the y-intercept.

- C, a are all numbers
- $C=0 \Rightarrow f(x) = 0 \cdot a^x = 0$, $a=1 \Rightarrow f(x) = C \cdot 2^x = C$,
  $a<0$ "$\Rightarrow$" $f(x) = C \cdot a^x$, $3 \cdot (-2)^{\frac{1}{2}} = 3\sqrt{-2}$ X

$C>0$: ① $a>1$, $f(x)$ growth

② $0<a<1$, $f(x)$ decay

$C<0$: reverse ① and ②

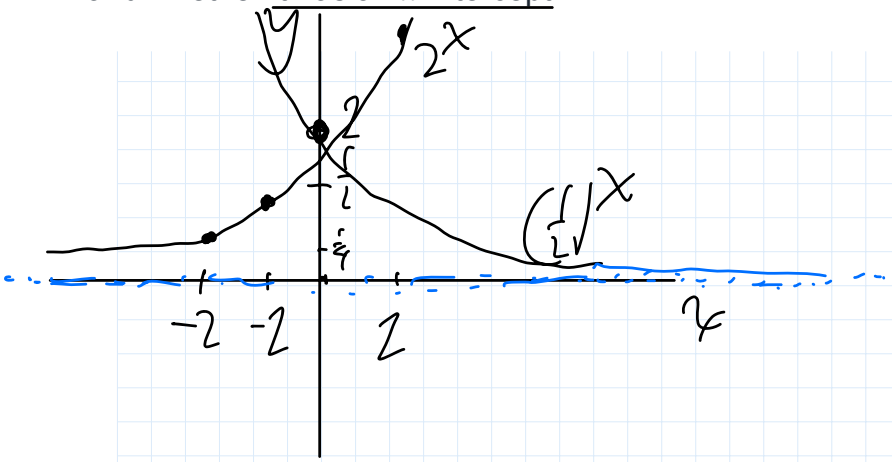
x	-2	-1	0	1	2
$y=2^x$	$2^{-2}=\frac{1}{2^2}=\frac{1}{4}$	$2^{-1}=\frac{1}{2}$	1	2	4
$y=\left(\frac{1}{2}\right)^x$	$\left(\frac{1}{2}\right)^{-2}=\frac{1}{2^{-2}}=\frac{1}{\frac{1}{2^2}}=1\cdot\frac{2^2}{1}=4$	$\left(\frac{1}{2}\right)^{-1}=2$	1	$\frac{1}{2}$	$\frac{1}{4}$

Sketch Growth and Decay

$x \to \infty$ $f(x) \to ?$

$x \to -\infty$

Use the table from the previous page to sketch $f(x) = 2^x$ and $g(x) = \left(\frac{1}{2}\right)^x$. For each function state the domain, range, horizontal asymptote, and whether it has an x-intercept.



Domain for $f(x) = 2^x$: $(-\infty, \infty)$

range ..... : $(0, \infty)$

D: $\left(\frac{1}{2}\right)^x$, $(-\infty, \infty)$

R: $\left(\frac{1}{2}\right)^x$, $(0, \infty)$

$x = 0$ is H.A.

$\lim_{x \to \infty} \left(\frac{1}{2}\right)^x = 0$

$\lim_{x \to -\infty} (2)^x = 0$

if $f(x)$ have an x-intercept

then $f(x) = 0$, solve for x.

but $2^x \neq 0$ there is no

x-intercept. Same $\left(\frac{1}{2}\right)^x$

Solve by Using the Same Base

For $a > 0$ and $a \neq 1$, the equation $a^u = a^v$ implies $u = v$. Solve by rewriting both sides with the same base:

(a) $2^{3x} = 16$ (b) $3^{2x-1} = 9$.

Check your answers in the original equations.

(a) $2^{3x} = 16$, want to find $a^u = a^v$

$2^{3x} = 4 \cdot 4 = 2^2 \cdot 2^2 = 2^4$

$\Downarrow$ $u = v$

$2^{3x} = 2^4$

$\Downarrow$

$3x = 4$

$x = \frac{4}{3}$

(b) $3^{2x-2} = 9$

$3^{2x-2} = 3^2$

$\Downarrow$

$2x - 2 = 2$

$2x - 3 = 0$

$x = \frac{3}{2}$

3^{2x-2} plug in $x = \frac{3}{2}$ $3^{2\cdot\frac{3}{2}-2} = 3^{3-2} = 3^2 = 9$ ✓

The Natural Base and a Quadratic Exponent

The constant $e \approx 2.71828$ is a common exponential base; it is not a variable. Solve and check:

(a) $e^{2x} = e^{x+3}$ (b) $e^{x^2} = e^{3x+4}$.

The same-base rule applies to e just as it applies to 2 or 3.

(a). $e^{2x} = e^{x+3}$ $\quad a^u = a^v \Rightarrow u = v$

$2x = x+3$

$x = 3$

(b). $e^{x^2} = e^{3x+4}$

$x^2 = 3x+4$

$x^2 - 3x - 4 = 0$

$(x-4)(x+1) = 0$

$x = 4$, $x = -1$.